

Vision–Language Models Read Their Own Attention at the Wrong Depth

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Abstract

Vision–language models decide *what to look at* — which visual tokens to keep, where to crop — by reading their own attention. We show the field reads it at the wrong depth. Question-conditioned localisation does not exist in the vision encoder (at chance on four architectures), is too weak at early language-model layers to be usable, and becomes reliable only in the upper half of the stack. The practical cost is large and, to our knowledge, unreported: ranking visual tokens for pruning by layer-2 attention — the default in a widely used method — is **worse than ranking them at random** on both models we test, losing 22.0pp and 13.1pp against no pruning at all. Reading late layers instead discards **90% of visual tokens at no measurable cost** (−0.5pp and −1.6pp), a margin of +24.1pp and +11.5pp over the early-layer default, with no training. The same defect appears in crop placement: replacing the standard block-mean read-out with a learned one improves evidence localisation on three of four architectures spanning two model families, and question answering by +12.0pp and +10.5pp over an unmodified model. A logit-lens analysis shows why cropping helps at all: it does not improve reasoning but restores an answer-formation step that appears abruptly at a single layer, and only when the crop actually delivers the evidence. We report what does not work with equal weight, including seven internal interventions that all fail and five of our own claims rejected by cross-architecture replication.

1 Introduction

A vision–language model (VLM) turns an image into a few hundred visual tokens. When the question concerns something small, the evidence occupies a fraction of one token and the model cannot answer: below roughly 0.15 merged tokens of target extent we measure 19.0% accuracy — *below the 25% chance level of a four-way question* — while the identical question asked of the resolved region is answered correctly 96–100% of the time.

Two large families of methods respond to this, and both make a decision from the model’s attention. Token pruning discards visual tokens judged unimportant. Crop-and-re-encode methods select a region and look again. **Both read attention, and the depth at which they read it has been treated as an implementation detail.** It is not.

Our contributions:

- **Pruning at the wrong depth is worse than not ranking at all.** Ranking visual tokens by layer-2 attention scores *below random selection* on two architectures. A late read-out at the same budget retains the accuracy of using every token while discarding 90% of them (§4).
- **The standard read-out averages away its own signal.** Collapsing a block of layers into one map and taking the maximum is worse than a learned combination on three of four architectures, two model families (§3).
- **A mechanism, with a causal control.** Cropping does not improve reasoning: cropped and uncropped trajectories are indistinguishable for twenty layers and then separate abruptly — and the separation appears *only* on items where the crop actually contains the evidence (§5).
- **An audited account of failure.** Seven internal interventions that would have been free if they worked, all null; and five claims of our own rejected when tested on a second architecture (§6, Appendix B).

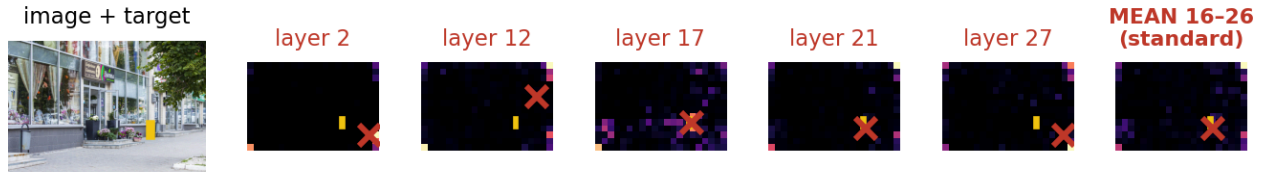


Figure 1: The layers disagree, and the standard read-out averages the disagreement away. One V*Bench item; the yellow box is the queried object. Each panel is attention over image positions at one layer; $\times$ marks its maximum (green inside the target, red outside). Individual layers disagree sharply. The rightmost panel is the read-out used in practice — the mean over layers 16–26 — whose maximum lands outside the target.

2 Setup

All experiments use V*Bench ($n = 191$ four-way questions with ground-truth boxes) and, for transfer, HR-Bench 4k ($n = 800$, CircularEval, 4032² images). Models: Qwen3-VL-2B, Qwen2-VL-7B, LLaVA-OneVision-7B, LLaVA-NeXT-7B. **Token counts are always measured** from the processor’s realised grid, never computed; any arm drifting more than 10% from its target voids its contrast. Every learned component is fitted **out-of-fold with folds grouped by item**. Appendix A gives the full protocol.

3 The read-out defect

The standard way to locate evidence in a VLM is to average attention over a block of layers and take the argmax. Figure 1 shows why this is lossy: layers disagree about where the target is, and a mean cannot represent disagreement.

Table 1: Evidence localisation, four architectures, two families. “Learned” is a ridge reweighting of per-layer maps, fitted out-of-fold. Chance is $\approx 2.2\%$ throughout.

model	block mean	learned	Δ	95% CI
Qwen3-VL-2B	39.3%	45.5%	+6.3pp	[+2.6, +10.5]
Qwen2-VL-7B	35.1%	43.5%	+8.4pp	[+3.1, +13.6]
LLaVA-OneVision-7B	25.1%	26.2%	+1.0pp	[−2.6, +5.2]
LLaVA-NeXT-7B	12.6%	18.8%	+6.3pp	[+2.1, +10.5]

Three of four clear zero; **LLaVA-OneVision is a null and we report it as one**. It is also the only model whose best single layer equals its block mean, i.e. whose layers agree — internally consistent, but a single data point we do not build on. Absolute localisation varies by a factor of three across models, which matters for anyone porting crop methods to LLaVA.

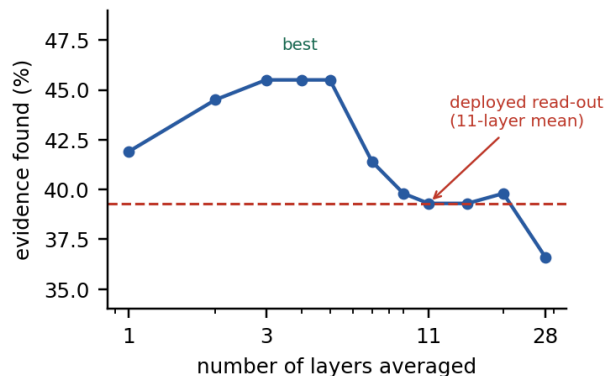


Figure 2: Averaging the k individually-best layers. More is worse past $k \approx 3$, and the deployed 11-layer mean sits well down the curve.

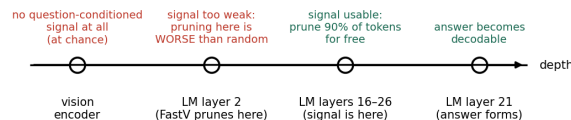


Figure 3: Where the question-conditioned signal lives, and what it costs to read it too early.

4 Pruning at the wrong depth

Token pruning ranks visual tokens by attention and drops the low-ranked ones. The widely used default reads that attention at layer 2. Figure 4 shows the consequence. Every arm prunes the same number of tokens, so cost is matched by construction.

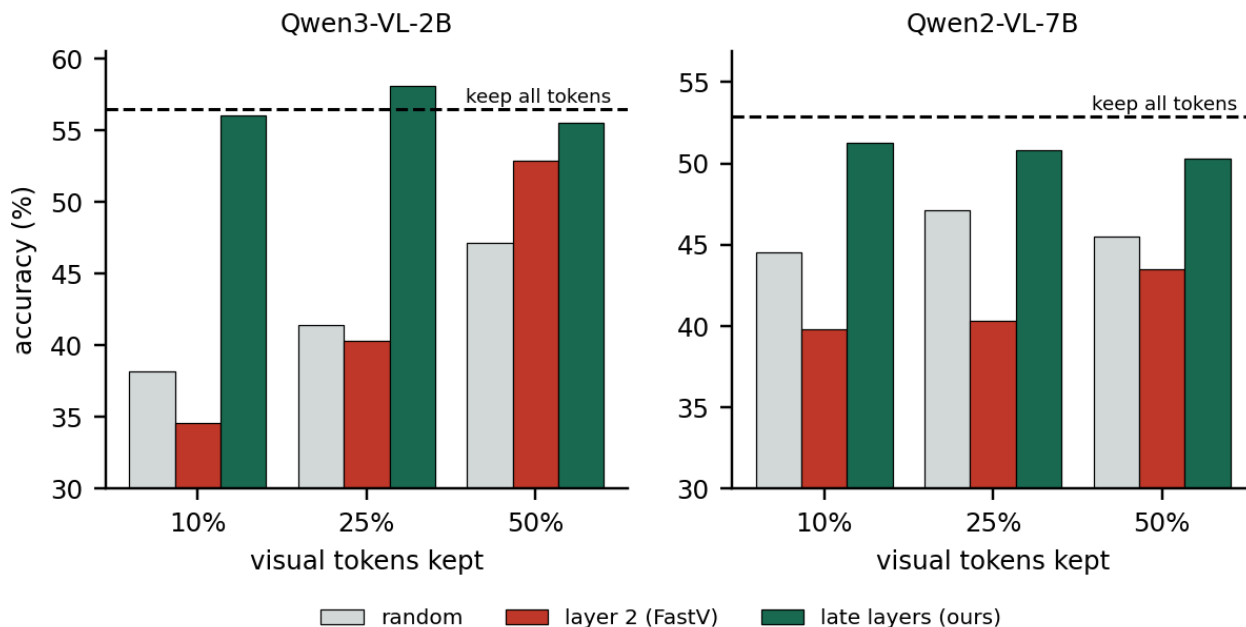


Figure 4: Ranking at layer 2 is worse than ranking at random. Dashed line: keeping every token. A late read-out discards 90% of visual tokens at no measurable cost, on both models.

At 10% keep, the late read-out is within noise of using every token (-0.5pp $[-5.8, +4.7]$ and -1.6pp $[-6.3, +3.1]$), while layer-2 ranking loses 22.0pp $[-29.8, -14.7]$ and 13.1pp respectively and falls *below random selection* on both models (-3.7pp , -4.7pp). The margin is $+24.1\text{pp}$ $[+16.2, +31.9]$ and $+11.5\text{pp}$ $[+4.2, +18.8]$.

No learning is required. A plain average of late layers matches a trained signed combination at every operating point ($+2.6$, -1.0 , $+1.0\text{pp}$, all null). The recommendation is a one-line change of which layer index is read.

A prediction that failed

We pre-registered that the penalty would be *larger* on Qwen2-VL, whose early layers rank the target worse. It is *smaller* (-13.1 vs -21.9 pp). Layer ranking quality therefore does not predict pruning damage: the claim replicates, its causal story does not, and we state them separately rather than let one carry the other.

5 Mechanism: what cropping actually does

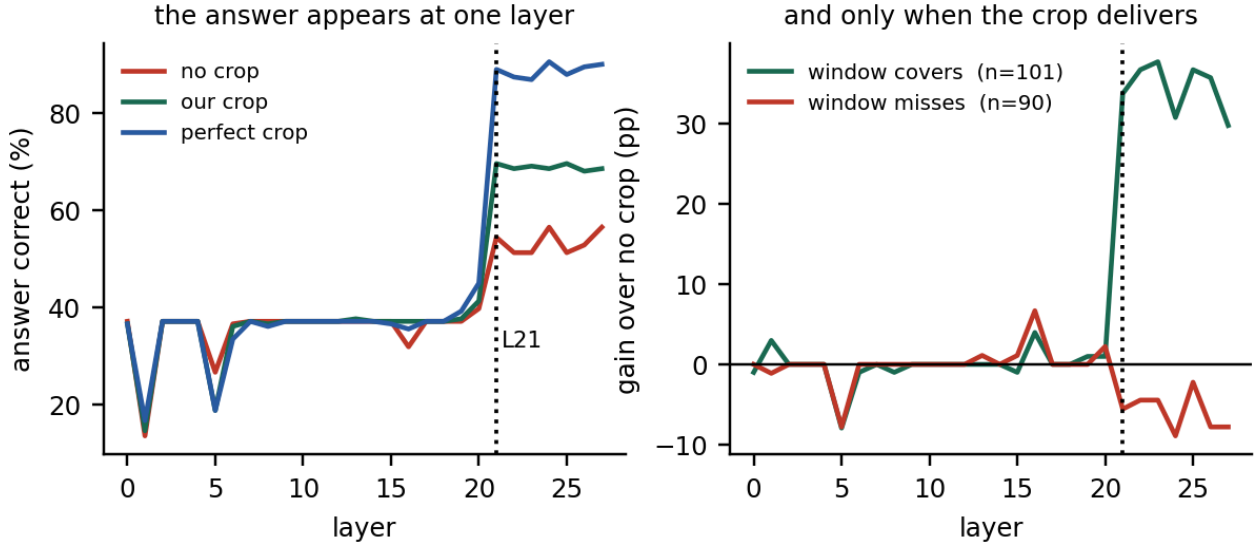


Figure 5: Cropping restores an answer-formation step; it does not improve reasoning. Left: the answer read out at every layer. Right: the same gain split by whether the crop contained the evidence.

Reading the model's answer at every layer (Figure 5) separates two explanations. A method that improved reasoning would pull ahead early and widen. Instead the cropped and uncropped arms are *indistinguishable through L20*, then separate by $+15.3$ pp at L21 and stay separated. The failing arms jump at L2 — the answer prior forming — and never improve: the uncropped arm's best score across all 28 layers *equals* its final answer, so there is no depth at which the answer was available and lost.

The right panel is the causal control. Splitting the *same* arm by whether its window contained the evidence: $+29.7$ pp $[+19.8, +39.6]$ where it covers, -7.9 pp $[-18.0, +2.2]$ where it misses — a 37.6 pp swing. The step appears only when the intervention delivers.

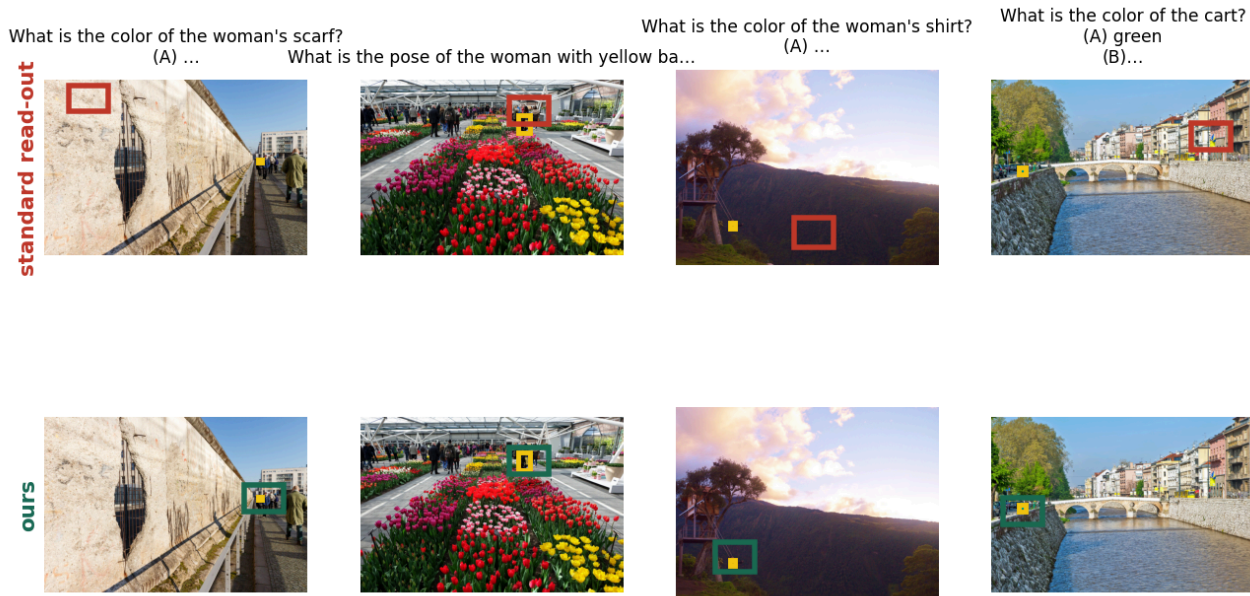


Figure 6: Items where the standard read-out misses and a learned one succeeds. Yellow: queried object. Red/green: the crop each read-out selects.

6 What does not work

Seven internal interventions, each free if it worked, are all null: suppressing the serialisation sink, amplifying attention on the target, residual-stream steering, layer read-out selection, DoLa/DeCo, component ablation, and contrastive decoding *steered to the ground-truth region* (ceiling +2.6pp [−0.5, +5.8]). The last is decisive because it aimed perfectly and verified the target is causally live — masking the evidence region shifts logits 3.32× more than masking a random region and costs 10.5pp — and still could not convert it.

The model looks in the right place; the right place is load-bearing; it still cannot answer, because no operation on the residual stream can supply information the encoder never wrote.

Four attempts to *regress a continuous geometric quantity* from internals also failed (budget from a perfect size oracle; budget from any free signal; window size from target size; attention-space reallocation). Ranking positions is learnable from internals; regressing scale is not.

7 Limitations

Our crop-placement method improves over an unmodified model on two architectures (+12.0pp [+4.2, +19.9], +10.5pp [+2.6, +18.3]) but **we have not shown it beats simply spending the same compute on a larger uniform image** (+4.7pp [−3.7, +13.1] and +3.1pp [−5.2, +11.0]). We report this rather than restricting to the stratum where it clears. Pruning baselines are reimplemented on a common backbone, not run from released checkpoints. The mechanism of §5 is verified on one model. Appendix B tags every claim by how many models support it.

Appendix A Protocol

Budgets are calibrated by iterating on the processor's own realised grid and are *measured*, never computed. Models load in bfloat16: loading a bf16 checkpoint in fp16 produced all-NaN logits whose argmax fell through to index 0, which

read as a clean negative result until caught. Learned components use grouped k -fold so an item's ~300 cells never straddle train and test. Bootstrap CIs are paired, 10,000 resamples. The logit lens takes intermediate layers as `lm_head(norm(h_i))` but the **final** layer from `model.logits`: re-normalising an already-normalised final state gives reconstruction error 23.47 against 0.06 and corrupts only that layer. Both paths are computed and asserted to disagree, so the bug cannot return silently. An executable suite of 84 invariants encodes every retraction below, so a claim the data stopped supporting fails a test rather than surviving in prose.

Appendix B Replication ledger

Table 2: Every claim by supporting models. A claim enters the paper only from the top block.

claim	models	status
Averaging dilutes the read-out	3 of 4, 2 families	survived
Anti-correlated layers exist	4	survived
Layer-2 pruning below random	2	survived
Learned read-out improves proposals	2	survived
Crop method beats unmodified model	2	survived
Encoding cliff	2	survived
Serialisation sink	4	survived
“The <i>final</i> layer is anti-correlated”	1 of 4	rejected
“Signed contrast is the mechanism”	1 of 2 tasks	rejected
“No single layer is a good localiser”	false on Qwen2-VL	rejected
Rank-only multi-crop	−3.1pp vs +5.5pp predicted	rejected
Crop method beats equal-compute baseline	0 of 2	rejected
Answer formation at a single layer	1	provisional
Zero-shot transfer to HR-Bench	1	provisional
Evidence region causally live	1	provisional
Seven internal nulls	1	provisional

Appendix C Corrections made during this work

A double-normalised logit lens manufactured a “free read-out gain” and a favourable baseline comparison; it was caught because a second code path computed the same quantity and disagreed. A transfer result was called negative from a category-ordered prefix and reversed at full n . Two mechanisms were proposed for a failure that turned out not to exist, and both were refuted by our own data. A capture fraction was quoted against a restricted rather than true ceiling (65% $\rightarrow$ 27.6%). An AUROC was computed on an unmasked map (0.576 $\rightarrow$ 0.745). In-sample “best layer” selection inverted a conclusion until fold-validated (41.9% $\rightarrow$ 36.1%, below the baseline it appeared to beat).